A Rank-Weighted Coloring Polynomial and the Partial-Dual Genus Polynomial

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Abstract

In this note, we organize Cheng's higher state sums for the graph-theoretic extension of the partial-dual genus polynomial into a rank-weighted coloring polynomial. We show that its binomial-basis coefficients recover the higher state sums, while its two-color specialization is Cheng's graph partial-dual genus polynomial and its specialization at $z = 0$ is the chromatic polynomial.

We also show that the linear coefficient in the color variable is the adjacency-rank character evaluated on the primitive projection in the graph Hopf algebra. This gives a direct expression for the partial-dual genus polynomial on the primitive projection and answers the evaluation problem posed by Cheng. Finally, we prove that the rank-weighted coloring polynomial satisfies the graph four-term relation, and hence that each of Cheng's higher state sums is a graph 4-invariant. [1]

1 Introduction

Partial duality for ribbon graphs with respect to a subset of edges was introduced by Chmutov [2]. Gross, Mansour, and Tucker subsequently introduced partial-dual genus polynomials, including the partial-dual Euler-genus polynomial, which records the Euler genera of all partial duals of a ribbon graph [3]. For orientable ribbon graphs, the Euler-genus polynomial is obtained from the orientable-genus polynomial by the substitution $z \mapsto z^2$. Chmutov proved that the orientable-genus polynomial, regarded as a function on chord diagrams, satisfies the four-term relation and hence defines a weight system [4]. Cheng then obtained an $\mathbb{F}_2$-rank formula for the partial-dual Euler-genus polynomial of the intersection graph of a chord diagram and used this formula to define an extension to all finite simple graphs; he also proved the graph four-term relation for this extension [1]. Following Cheng, the partial-dual genus polynomial considered below is this graph-theoretic extension.

In Section 4.2 of [1], Cheng considered the adjacency-rank character, higher state sums over nonempty partitions of the vertex set, and the problem of determining the partial-dual genus polynomial on the primitive projection in the graph Hopf algebra. This question is parallel to a result of Dogra and Lando, who showed that their skew characteristic polynomial has particularly simple behavior on primitive elements [5].

In this note, we package Cheng's higher state sums into a rank-weighted coloring polynomial. Unlike Cheng's refined partial-dual genus polynomial, which refines a single subset and its complement, the new polynomial sums over colorings and hence over partitions with an arbitrary number of color classes. Its binomial-basis coefficients are precisely Cheng's higher state sums. We show that its linear color coefficient is the adjacency-rank character evaluated on the primitive projection, giving a direct formula for the partial-dual genus polynomial there. We also prove that the rank-weighted

coloring polynomial satisfies the graph four-term relation; coefficient comparison then gives the four-term relation for every higher state sum.

2 Preliminaries

Throughout, graphs are finite and simple, and all adjacency matrices and their ranks are considered over $\mathbb{F}_2$. We denote by $\mathbf{1}$ the graph with no vertices. Let G be a graph and let M_G denote its adjacency matrix. For $A \subseteq V(G)$, let $G[A]$ denote the induced subgraph of G on A, and let $M_A(G)$ denote the principal submatrix of M_G indexed by A; equivalently, it is the adjacency matrix of $G[A]$ with the vertex order inherited from M_G. When the ambient graph is clear, we abbreviate $M_A(G)$ to M_A. Thus $G[\varnothing] = \mathbf{1}$ and $\operatorname{rank}(M_\varnothing) = 0$. Write $A^c = V(G) \setminus A$.

For a positive integer m, write $[m] = \{1, \dots, m\}$. An m-coloring is a map $c : V(G) \to [m]$. It is proper if $c(u) \neq c(v)$ for every edge $uv \in E(G)$. We write $\chi_G(t)$ for the chromatic polynomial, so $\chi_G(m)$ is the number of proper m-colorings.

For a polynomial $F(t, z)$ and integers $i, j \geq 0$, write $[t^i]F(t, z)$ and $[z^j]F(t, z)$ for the coefficients of t^i and z^j, respectively, with the other variable retained. In particular, $[t]F(t, z) = [t^1]F(t, z)$.

Definition 2.1 (Partial-dual genus polynomial)**.** Following Cheng's extension to finite simple graphs [1, Definition 3.2], define

$$^{\partial}\varepsilon_G(z) = \sum_{A \subseteq V(G)} z^{\operatorname{rank}(M_A)+\operatorname{rank}(M_{A^c})}.$$

For an intersection graph arising from a chord diagram, this agrees with the partial-dual Euler-genus polynomial of the associated ribbon graph. In particular, $^{\partial}\varepsilon_{\mathbf{1}}(z) = 1$.

3 The rank-weighted coloring polynomial

Definition 3.1 (Rank-weighted coloring polynomial)**.** Let m be a positive integer. For a coloring $c : V(G) \to [m]$, define its rank weight by

$$\operatorname{rw}(c) := \sum_{i=1}^{m} \operatorname{rank}\big(M_{c^{-1}(i)}\big).$$

The rank-weighted coloring state sum is

$$\mathcal{R}_G(m, z) := \sum_{c : V(G) \to [m]} z^{\operatorname{rw}(c)}.$$

The variable m records the number of colors, and the exponent of z records total adjacency rank. Proposition 3.4 shows that this state sum extends polynomially in the color variable.

For the empty graph, set $\mathcal{R}_{\mathbf{1}}(t, z) = 1$; this agrees with the state sum $\mathcal{R}_{\mathbf{1}}(m, z) = 1$ for every positive integer m.

We next recall the higher state sums considered by Cheng in Section 4.2 of [1]. We use the ordered version of the indexed partition sum. For $k \geq 1$, let $\Pi_k(V(G))$ denote the set of unordered partitions of $V(G)$ into k nonempty blocks.

Definition 3.2 (Higher partial-dual genus state sums)**.** For every $k \geq 1$, define

$$\partial\varepsilon_G^{(k)}(z) := \sum_{\substack{A_1 \sqcup \cdots \sqcup A_k = V(G) \\ A_i \neq \varnothing}} z^{\sum_{i=1}^{k} \operatorname{rank}(M_{A_i})},$$

where $(A_1, \ldots, A_k)$ is an ordered k-tuple of nonempty blocks. The sum is understood to be 0 if $k > |V(G)|$.

Remark 3.3. If the blocks are regarded as unordered, then

$$\partial\varepsilon_G^{(k)}(z) = k! \sum_{\mathcal{P} \in \Pi_k(V(G))} z^{\sum_{B \in \mathcal{P}} \operatorname{rank}(M_B)}.$$

For every nonempty graph G,

$$\partial\varepsilon_G(z) = 2\partial\varepsilon_G^{(1)}(z) + \partial\varepsilon_G^{(2)}(z).$$

Indeed, the two one-color contributions to $\mathcal{R}_G(2, z)$ yield $2\partial\varepsilon_G^{(1)}(z)$, and the colorings using both colors yield $\partial\varepsilon_G^{(2)}(z)$.

Proposition 3.4 (Set-partition and binomial-basis expansions)**.** *Let G have $n \geq 1$ vertices. Then the state sum $m \mapsto \mathcal{R}_G(m, z)$ agrees on the positive integers with a unique polynomial $\mathcal{R}_G(t, z)$.*

For $1 \leq k \leq n$, write

$$(t)_k := t(t-1)\cdots(t-k+1).$$

Then

$$\mathcal{R}_G(t, z) = \sum_{k=1}^{n} (t)_k \sum_{\mathcal{P} \in \Pi_k(V(G))} z^{\sum_{B \in \mathcal{P}} \operatorname{rank}(M_B)}.$$

In particular, $\mathcal{R}_G(t, z) \in \mathbb{Z}[t, z]$. Moreover,

$$\partial\varepsilon_G^{(k)}(z) = k! \sum_{\mathcal{P} \in \Pi_k(V(G))} z^{\sum_{B \in \mathcal{P}} \operatorname{rank}(M_B)},$$

and hence

$$\mathcal{R}_G(t, z) = \sum_{k=1}^{n} \binom{t}{k} \partial\varepsilon_G^{(k)}(z).$$

Proof. Fix $m \geq 1$ and partition the colorings $c : V(G) \to [m]$ according to the number k of colors that actually occur. Suppose that c uses exactly k colors, and write its image as

$$\{s_1 < \cdots < s_k\} \subseteq [m].$$

Set $A_j = c^{-1}(s_j)$. Then

$$A_1 \sqcup \cdots \sqcup A_k = V(G), \qquad A_j \neq \varnothing,$$

and the unused colors contribute rank zero. Hence

$$\operatorname{rw}(c) = \sum_{j=1}^{k} \operatorname{rank}(M_{A_j}).$$

Conversely, a choice of a k-element subset $\{s_1 < \cdots < s_k\} \subseteq [m]$ together with an ordered partition $(A_1, \ldots, A_k)$ determines a unique coloring by assigning the color s_j to the vertices of A_j.

This correspondence preserves the rank weight. There are $\binom{m}{k}$ choices of the set of used colors, so the total contribution of colorings using exactly k colors is

$$\binom{m}{k}\partial_{\varepsilon_G}^{(k)}(z).$$

Summing over $1 \le k \le n$ gives, for every positive integer m,

$$\mathcal{R}_G(m,z) = \sum_{k=1}^{n} \binom{m}{k}\partial_{\varepsilon_G}^{(k)}(z).$$

The right-hand side lies in $\mathbb{Q}[z][m]$, so it defines a polynomial $\mathcal{R}_G(t,z)$ agreeing with the coloring state sum at every positive integer. Thus

$$\mathcal{R}_G(t,z) = \sum_{k=1}^{n} \binom{t}{k}\partial_{\varepsilon_G}^{(k)}(z).$$

This polynomial is unique: a nonzero polynomial over the integral domain $\mathbb{Q}[z]$ has at most its degree many roots, so two elements of $\mathbb{Q}[z][t]$ that agree at every positive integer are identical.

On the other hand, every unordered partition $\mathcal{P} \in \Pi_k(V(G))$ has exactly $k!$ orderings, and the rank weight is unchanged by permuting its blocks. Hence

$$\partial_{\varepsilon_G}^{(k)}(z) = k! \sum_{\mathcal{P}\in\Pi_k(V(G))} z^{\sum_{B\in\mathcal{P}} \operatorname{rank}(M_B)}.$$

Since $(t)_k = k!\binom{t}{k}$, substitution into the preceding identity gives

$$\mathcal{R}_G(t,z) = \sum_{k=1}^{n} (t)_k \sum_{\mathcal{P}\in\Pi_k(V(G))} z^{\sum_{B\in\mathcal{P}} \operatorname{rank}(M_B)}.$$

Because each $(t)_k$ lies in $\mathbb{Z}[t]$, this also shows that $\mathcal{R}_G(t,z) \in \mathbb{Z}[t,z]$. □

Theorem 3.5 (Basic specializations)**.** *Let G be a finite simple graph on n vertices. Then*

$$\begin{aligned}
\mathcal{R}_G(t,1) &= t^n,\\
\mathcal{R}_G(t,0) &= \chi_G(t),\\
\mathcal{R}_G(1,z) &= z^{\operatorname{rank}(M_G)},\\
\mathcal{R}_G(2,z) &= \partial\varepsilon_G(z).
\end{aligned}$$

Proof. Fix a positive integer m. If $z = 1$, every coloring has weight 1, and there are m^n maps $V(G) \to [m]$. Thus

$$\mathcal{R}_G(m,1) = m^n.$$

Since both sides are polynomials in m, this gives $\mathcal{R}_G(t,1) = t^n$.

For $z = 0$, the constant term of $\mathcal{R}_G(m,z)$ receives a contribution from a coloring precisely when every color class has adjacency rank zero. For a simple graph, an adjacency matrix has rank zero if and only if it is the zero matrix, equivalently if and only if the corresponding induced subgraph has no edges. Thus the contributing colorings are exactly the proper colorings, and

$$\mathcal{R}_G(m,0) = \chi_G(m).$$

Polynomiality gives $\mathcal{R}_G(t,0) = \chi_G(t)$.

For $m = 1$, there is one coloring and its sole color class is $V(G)$. Hence

$$\mathcal{R}_G(1,z) = z^{\operatorname{rank}(M_G)}.$$

For $m = 2$, a coloring is uniquely determined by $A = c^{-1}(1)$, with the other color class equal to A^c. Therefore

$$\mathcal{R}_G(2,z) = \sum_{A \subseteq V(G)} z^{\operatorname{rank}(M_A)+\operatorname{rank}(M_{A^c})} = {}^{\partial}\varepsilon_G(z),$$

where the last equality follows from [1, Definition 3.2]. □

Proposition 3.6 (Multiplicativity)**.** *For finite simple graphs G and H,*

$$\mathcal{R}_{G \sqcup H}(t,z) = \mathcal{R}_G(t,z)\mathcal{R}_H(t,z).$$

Proof. Fix $m \geq 1$. A coloring of $G \sqcup H$ is uniquely a pair of colorings $c_G : V(G) \to [m]$ and $c_H : V(H) \to [m]$. For each color i, the induced subgraph on the ith color class is

$$G[c_G^{-1}(i)] \sqcup H[c_H^{-1}(i)],$$

whose adjacency matrix is block diagonal. Hence its rank is

$$\operatorname{rank}\big(M_{c_G^{-1}(i)}(G)\big) + \operatorname{rank}\big(M_{c_H^{-1}(i)}(H)\big).$$

Thus the rank weights add, and hence the monomial weight $z^{\operatorname{rw}(c)}$ is the product of the two monomial weights. Summing independently over c_G and c_H gives

$$\mathcal{R}_{G \sqcup H}(m,z) = \mathcal{R}_G(m,z)\mathcal{R}_H(m,z).$$

The identity for t follows by polynomiality. □

Example 3.7. For the one-vertex graph K_1, every color class has rank zero, so

$$\mathcal{R}_{K_1}(t,z) = t.$$

For K_2, a coloring in which the two vertices have different colors has weight 1, while a monochromatic coloring has weight z^2. Hence

$$\mathcal{R}_{K_2}(t,z) = t(t-1) + tz^2 = t(t-1+z^2).$$

At $t = 2$ this gives

$${}^{\partial}\varepsilon_{K_2}(z) = 2 + 2z^2.$$

Example 3.8. For K_4, the adjacency matrix of an induced K_s has rank s when s is even and rank $s-1$ when s is odd. Sorting set partitions of $V(K_4)$ by their block-size types gives

$$\mathcal{R}_{K_4}(t,z) = (t)_4 + \big(4(t)_2 + 6(t)_3\big)z^2 + \big(t + 3(t)_2\big)z^4.$$

At $z = 1$, this becomes t^4; at $z = 0$, it becomes $(t)_4 = \chi_{K_4}(t)$; and at $t = 2$, it gives

$${}^{\partial}\varepsilon_{K_4}(z) = 8z^2 + 8z^4.$$

3.1 Rank defect and coefficient structure

The exponent of z may equivalently be expressed through the adjacency nullities of the color classes.

Definition 3.9. Let $c : V(G) \to [m]$ be a coloring and let $C_i = c^{-1}(i)$. Define its total adjacency nullity by

$$\delta(c) := \sum_{i=1}^{m} \big(|C_i| - \operatorname{rank}(M_{C_i})\big).$$

Proposition 3.10 (Nullity-defect interpretation)**.** *Let G have n vertices. Since $\operatorname{rank}(M_A) \le |A|$ for every $A \subseteq V(G)$, the degree of $\mathcal{R}_G(t,z)$ in z is at most n. Write*

$$\mathcal{R}_G(t,z) = \sum_{d=0}^{n} D_{G,d}(t) z^{n-d}.$$

For every positive integer m,

$$D_{G,d}(m) = \#\{c : V(G) \to [m] : \delta(c) = d\}.$$

In particular,

$$D_{G,n}(t) = \chi_G(t).$$

Moreover,

$$D_{G,d}(t) = 0 \qquad \textit{unless} \qquad n - d \equiv 0 \pmod 2.$$

Proof. Let $C_i = c^{-1}(i)$. Since the color classes partition $V(G)$,

$$\sum_{i=1}^{m} |C_i| = n.$$

Therefore

$$\delta(c) = \sum_{i=1}^{m} \big(|C_i| - \operatorname{rank}(M_{C_i})\big) = n - \sum_{i=1}^{m} \operatorname{rank}(M_{C_i}).$$

Equivalently,

$$\sum_{i=1}^{m} \operatorname{rank}(M_{C_i}) = n - \delta(c).$$

Thus c contributes to the coefficient of z^{n-d} exactly when $\delta(c) = d$, proving the first assertion. The identity $D_{G,n}(t) = \chi_G(t)$ is the constant-term case and follows from Theorem 3.5.

Finally, the adjacency matrix of a simple graph is symmetric with zero diagonal. Over $\mathbb{F}_2$, it is alternating: for every vector v, one has $v^T M v = 0$, since the off-diagonal terms occur in equal pairs and the diagonal is zero. Every alternating matrix has even rank. Hence, if $n - d$ is odd, then no coloring can satisfy $\delta(c) = d$. By the first assertion, $D_{G,d}(m) = 0$ for every positive integer m.

Since $D_{G,d}(t) \in \mathbb{Z}[t]$ vanishes at infinitely many integers, it is the zero polynomial. □

Remark 3.11. The coefficient $D_{G,0}(m) = [z^n]\mathcal{R}_G(m,z)$ counts colorings for which every nonempty color class has nonsingular adjacency matrix over $\mathbb{F}_2$. Thus the two extreme defect conditions are proper colorings and colorings by nonsingular induced adjacency blocks.

4 The graph Hopf algebra and the primitive projection

We use the graph bialgebra considered in Section 4.2 of [1]; see also [6]. Cheng works over $\mathbb{C}$, whereas we work over $\mathbb{Q}$. Let $\mathcal{H}$ be the $\mathbb{Q}$-vector space spanned by isomorphism classes of finite simple graphs. Its product and coproduct are

$$G \cdot H = G \sqcup H, \qquad \Delta(G) = \sum_{A \subseteq V(G)} G[A] \otimes G[A^c].$$

The empty graph $\mathbf{1}$ is the unit, and the counit sends $\mathbf{1}$ to 1 and every nonempty graph to 0. With the grading $\deg(G) = |V(G)|$, $\mathcal{H}$ is a connected graded commutative and cocommutative bialgebra, and hence a Hopf algebra.

Define the linear map

$$r_z : \mathcal{H} \longrightarrow \mathbb{Q}[z], \qquad r_z(G) = z^{\operatorname{rank}(M_G)}.$$

Since $r_z(\mathbf{1}) = 1$ and the adjacency matrix of a disjoint union is block diagonal,

$$r_z(G \sqcup H) = r_z(G) r_z(H),$$

so r_z is a character, that is, a unital algebra homomorphism. For linear maps $f, g : \mathcal{H} \to \mathbb{Q}[z]$, write

$$(f * g)(x) = m_{\mathbb{Q}[z]}(f \otimes g)\Delta(x)$$

for convolution, where $m_{\mathbb{Q}[z]}$ denotes multiplication in $\mathbb{Q}[z]$. For $m \geq 1$, write f^{*m} for the m-fold convolution power of f.

Lemma 4.1 (Colorings as convolution powers)**.** *For every finite simple graph G and every positive integer m,*

$$\mathcal{R}_G(m, z) = r_z^{*m}(G).$$

In particular,

$$\partial_{\varepsilon_G}(z) = (r_z * r_z)(G).$$

Proof. The $(m-1)$-fold iterated coproduct, with $\Delta^{(0)} = \mathrm{Id}_{\mathcal{H}}$, is

$$\Delta^{(m-1)}(G) = \sum_{A_1 \sqcup \cdots \sqcup A_m = V(G)} G[A_1] \otimes \cdots \otimes G[A_m],$$

where empty blocks are allowed. Such an ordered decomposition is equivalent to a map $V(G) \to [m]$. Applying r_z to the tensor factors and multiplying gives

$$\begin{aligned} r_z^{*m}(G) &= \sum_{A_1 \sqcup \cdots \sqcup A_m = V(G)} \prod_{i=1}^{m} z^{\operatorname{rank}(M_{A_i})} \\ &= \sum_{A_1 \sqcup \cdots \sqcup A_m = V(G)} z^{\sum_{i=1}^{m} \operatorname{rank}(M_{A_i})} \\ &= \mathcal{R}_G(m, z). \end{aligned}$$

Taking $m = 2$ and comparing with [1, Definition 3.2] gives the second identity. □

We next use the primitive projection appearing in Section 4.2 of [1]. An element $x \in \mathcal{H}$ is primitive if

$$\Delta(x) = \mathbf{1} \otimes x + x \otimes \mathbf{1}.$$

Following [1, Section 4.2], let $\pi : \mathcal{H} \to \mathcal{H}$ denote the primitive projection, with $\pi(\mathbf{1}) = 0$. If G is nonempty and $n = |V(G)|$, then

$$\begin{aligned} \pi(G) &= \sum_{k=1}^{n} (-1)^{k-1}(k-1)! \sum_{\mathcal{P} \in \Pi_k(V(G))} \prod_{B \in \mathcal{P}} G[B] \\ &= \sum_{k=1}^{n} \frac{(-1)^{k-1}}{k} \sum_{\substack{A_1 \sqcup \cdots \sqcup A_k = V(G) \\ A_i \neq \varnothing}} G[A_1] \cdots G[A_k]. \end{aligned} \tag{1}$$

The two expressions agree because every unordered partition into k blocks has $k!$ orderings. All products in (1) are products in $\mathcal{H}$, so $\pi(G)$ is generally a formal $\mathbb{Q}$-linear combination of graphs. In particular,

$$\Delta(\pi(G)) = \mathbf{1} \otimes \pi(G) + \pi(G) \otimes \mathbf{1}.$$

Theorem 4.2 (The linear color coefficient)**.** *For every nonempty finite simple graph G,*

$$[t]\mathcal{R}_G(t, z) = r_z(\pi(G)).$$

Equivalently,

$$r_z(\pi(G)) = \sum_{k=1}^{|V(G)|} \frac{(-1)^{k-1}}{k} \partial_{\varepsilon_G}^{(k)}(z). \tag{2}$$

Proof. Let $n = |V(G)|$. By Proposition 3.4,

$$\mathcal{R}_G(t, z) = \sum_{k=1}^{n} \binom{t}{k} \partial_{\varepsilon_G}^{(k)}(z).$$

For $k \geq 1$,

$$[t]\binom{t}{k} = \frac{(-1)^{k-1}(k-1)!}{k!} = \frac{(-1)^{k-1}}{k}.$$

Hence

$$[t]\mathcal{R}_G(t, z) = \sum_{k=1}^{n} \frac{(-1)^{k-1}}{k} \partial_{\varepsilon_G}^{(k)}(z). \tag{3}$$

On the other hand, applying r_z to (1) and using multiplicativity gives

$$\begin{aligned} r_z(\pi(G)) &= \sum_{k=1}^{n} \frac{(-1)^{k-1}}{k} \sum_{\substack{A_1 \sqcup \cdots \sqcup A_k = V(G) \\ A_i \neq \varnothing}} r_z\big(G[A_1] \cdots G[A_k]\big) \\ &= \sum_{k=1}^{n} \frac{(-1)^{k-1}}{k} \sum_{\substack{A_1 \sqcup \cdots \sqcup A_k = V(G) \\ A_i \neq \varnothing}} z^{\sum_{i=1}^{k} \operatorname{rank}(M_{A_i})} \\ &= \sum_{k=1}^{n} \frac{(-1)^{k-1}}{k} \partial_{\varepsilon_G}^{(k)}(z). \end{aligned}$$

Comparison with (3) proves the theorem. □

Corollary 4.3 (Partial-dual genus evaluation on the primitive projection)**.** *Extend the map $G \mapsto {}^{\partial}\varepsilon_G(z)$ linearly to a map ${}^{\partial}\varepsilon : \mathcal{H} \to \mathbb{Q}[z]$, and retain the notation ${}^{\partial}\varepsilon_X(z)$ for its value at $X \in \mathcal{H}$. For every nonempty finite simple graph G,*

$$ {}^{\partial}\varepsilon_{\pi(G)}(z) = 2[t]\mathcal{R}_G(t,z). $$

Equivalently,

$$ {}^{\partial}\varepsilon_{\pi(G)}(z) = 2\sum_{k=1}^{|V(G)|} \frac{(-1)^{k-1}}{k} {}^{\partial}\varepsilon_G^{(k)}(z). $$

Proof. By Lemma 4.1, the linear maps ${}^{\partial}\varepsilon$ and $r_z * r_z$ agree on the graph basis of $\mathcal{H}$, and hence agree on all of $\mathcal{H}$. Since $\pi(G)$ is primitive,

$$\begin{aligned} {}^{\partial}\varepsilon_{\pi(G)}(z) &= (r_z * r_z)(\pi(G)) \\ &= m_{\mathbb{Q}[z]}(r_z \otimes r_z)\Delta(\pi(G)) \\ &= r_z(\mathbf{1})r_z(\pi(G)) + r_z(\pi(G))r_z(\mathbf{1}). \end{aligned}$$

Since $r_z(\mathbf{1}) = 1$, this is $2r_z(\pi(G))$. Theorem 4.2 now gives the first formula, and (2) gives the second. □

Corollary 4.4 (Chromatic information in the primitive term)**.** *For every nonempty finite simple graph G,*

$$ [z^0]{}^{\partial}\varepsilon_{\pi(G)}(z) = 2[t]\chi_G(t). $$

Proof. By Corollary 4.3,

$$ [z^0]{}^{\partial}\varepsilon_{\pi(G)}(z) = 2[z^0][t]\mathcal{R}_G(t,z) = 2[t][z^0]\mathcal{R}_G(t,z). $$

Theorem 3.5 gives $[z^0]\mathcal{R}_G(t,z) = \chi_G(t)$, proving the claim. □

5 The four-term relation

We use Lando's graph four-term operations [6], in the convention used by Cheng [1, Section 3]. For a graph H and a vertex $v \in V(H)$, let $N_H(v)$ denote the set of neighbors of v in H. Let G be a simple graph and let $a, b \in V(G)$ be distinct. Let G'_{ab} be obtained by toggling the edge ab. Let $\widetilde{G}_{ab}$ be obtained by toggling the adjacency between a and every vertex of $N_G(b) \setminus \{a\}$. Since $N_{G'_{ab}}(b) \setminus \{a\} = N_G(b) \setminus \{a\}$, the two operations commute. We denote their common composition by

$$ \widetilde{G'_{ab}} = (\widetilde{G}_{ab})'_{ab}. $$

A graph invariant f with values in an abelian group is called a graph 4-invariant if

$$ f(G) - f(G'_{ab}) - f(\widetilde{G}_{ab}) + f(\widetilde{G'_{ab}}) = 0 $$

for every finite simple graph G and every pair of distinct vertices $a, b \in V(G)$.

Theorem 5.1 (Four-term relation)**.** *For every finite simple graph G and every pair of distinct vertices $a, b \in V(G)$,*

$$ \mathcal{R}_G(t,z) - \mathcal{R}_{G'_{ab}}(t,z) - \mathcal{R}_{\widetilde{G}_{ab}}(t,z) + \mathcal{R}_{\widetilde{G'_{ab}}}(t,z) = 0. $$

Thus $\mathcal{R}_G(t,z)$ is a graph 4-invariant.

Proof. Let

$$F_{a,b}(t,z) := \mathcal{R}_G(t,z) - \mathcal{R}_{G'_{ab}}(t,z) - \mathcal{R}_{\widetilde{G}_{ab}}(t,z) + \mathcal{R}_{\widetilde{G'_{ab}}}(t,z).$$

Since $F_{a,b}(t,z)$ is a polynomial in t with coefficients in $\mathbb{Q}[z]$, it suffices to prove

$$F_{a,b}(m,z) = 0$$

for every positive integer m.

Fix a coloring $c : V(G) \to [m]$. The same map c is also a coloring of the other three graphs. We show that its four contributions cancel.

First suppose $c(a) \neq c(b)$. The graphs G and G'_{ab} differ only in the edge ab, whose endpoints lie in different color classes. Hence every monochromatic induced subgraph is the same in these two graphs, so the weights of c in G and G'_{ab} are equal. The same argument applies to $\widetilde{G}_{ab}$ and $\widetilde{G'_{ab}}$, which also differ only in the edge ab. Thus the four contributions cancel.

Now suppose $c(a) = c(b)$, and let C be their common color class. Any edge changed by the tilde operation has one endpoint a. Therefore the only monochromatic adjacency matrix that can change is the one induced by C. Order the vertices of C as $a, b, C \setminus \{a,b\}$. Over $\mathbb{F}_2$ write

$$M_C(G) = \begin{pmatrix} 0 & \delta & x^T \\ \delta & 0 & y^T \\ x & y & B \end{pmatrix},$$

where $\delta \in \mathbb{F}_2$ records whether a and b are adjacent. The tilde operation toggles the adjacency between a and every neighbor of b in $C \setminus \{a,b\}$. Consequently

$$M_C(\widetilde{G}_{ab}) = \begin{pmatrix} 0 & \delta & (x+y)^T \\ \delta & 0 & y^T \\ x+y & y & B \end{pmatrix}.$$

This matrix is obtained from $M_C(G)$ by the elementary operations

$$R_a \leftarrow R_a + R_b, \qquad C_a \leftarrow C_a + C_b.$$

Both operations are invertible, hence

$$\operatorname{rank}\big(M_C(G)\big) = \operatorname{rank}\big(M_C(\widetilde{G}_{ab})\big).$$

The identical argument with δ replaced by $\delta + 1$ gives

$$\operatorname{rank}\big(M_C(G'_{ab})\big) = \operatorname{rank}\big(M_C(\widetilde{G'_{ab}})\big).$$

All other color classes have unchanged induced adjacency matrices. Therefore, the weight of c in G equals its weight in $\widetilde{G}_{ab}$, while its weight in G'_{ab} equals its weight in $\widetilde{G'_{ab}}$. Again the four contributions cancel.

Since the cancellation holds for every coloring c, summing over all colorings gives $F_{a,b}(m,z) = 0$ for every positive integer m. Hence $F_{a,b}(t,z) = 0$ identically. □

Remark 5.2. At $t = 2$, Theorem 5.1 recovers Cheng's four-term theorem for the partial-dual genus polynomial [1, Theorem 1.2]. The proof above shows that the same cancellation occurs coloring by coloring for an arbitrary number of colors.

Corollary 5.3. *For every $k \geq 1$, the map $G \mapsto {}^{\partial}\varepsilon_G^{(k)}(z)$ is a graph 4-invariant.*

Proof. Fix $k \geq 1$, a finite simple graph G, and distinct vertices $a, b \in V(G)$, and set $n = |V(G)|$. If $k > n$, then ${}^{\partial}\varepsilon_G^{(k)}(z)$ and the corresponding higher state sums of the other three graphs are zero by Definition 3.2, since all four graphs have vertex set $V(G)$. Assume now that $1 \leq k \leq n$. By Proposition 3.4 and Theorem 5.1,

$$\begin{aligned} 0 &= \mathcal{R}_G(t,z) - \mathcal{R}_{G'_{ab}}(t,z) - \mathcal{R}_{\widetilde{G}_{ab}}(t,z) + \mathcal{R}_{\widetilde{G'_{ab}}}(t,z) \\ &= \sum_{\ell=1}^{n} \binom{t}{\ell} \Big({}^{\partial}\varepsilon_G^{(\ell)}(z) - {}^{\partial}\varepsilon_{G'_{ab}}^{(\ell)}(z) - {}^{\partial}\varepsilon_{\widetilde{G}_{ab}}^{(\ell)}(z) + {}^{\partial}\varepsilon_{\widetilde{G'_{ab}}}^{(\ell)}(z) \Big). \end{aligned}$$

The polynomials $\binom{t}{1}, \ldots, \binom{t}{n}$ have distinct degrees in t and leading coefficients $1, 1/2!, \ldots, 1/n!$, so they are linearly independent over $\mathbb{Q}[z]$. Hence the coefficient of $\binom{t}{k}$ vanishes, which is precisely the four-term relation for ${}^{\partial}\varepsilon_G^{(k)}(z)$. □

6 Discussion

The rank-weighted coloring polynomial suggests two immediate directions.

1. Find structural descriptions and closed forms for the intermediate defect polynomials

$$D_{G,d}(t) = [z^{|V(G)|-d}]\mathcal{R}_G(t,z).$$

   The extreme case $d = |V(G)|$ is the chromatic polynomial, while $D_{G,0}(m)$ counts, for each positive integer m, colorings into nonsingular induced adjacency blocks.

2. Compare the information carried by $\mathcal{R}_G(t,z)$ with Cheng's refined partial-dual genus polynomial and with other rank-based graph invariants. In particular, determine whether natural graph families admit closed forms for the coefficients in either variable.

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